SMART MANUFACTURING AI

Visual Quality Inspection & Automated Defect Intelligence Platform

RESULT: FAIL

FAILED QUALITY AUDIT

Inspection ID:	INSP-1793371144-D1BC7C	**Timestamp:**	2026-10-30 20:09:07
Component Name:	Tractor engine housing	**Similarity Score (SSIM):**	99.04%
Total Defect Regions:	2	**Affected Surface Area:**	1.563%
High-Severity Defects:	1	**Processing Duration:**	2570.8 ms

Visual Evidence & Comparative Computer Vision Analysis

Figure A: Reference (Good Component)



Figure B: Inspected Component

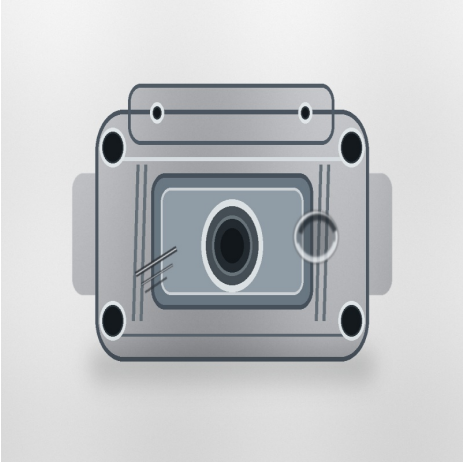


Figure C: SSIM Difference Heatmap

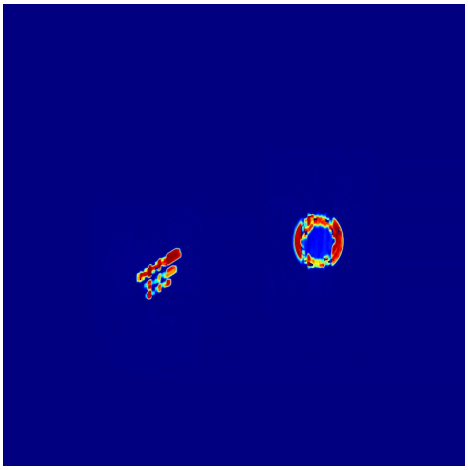
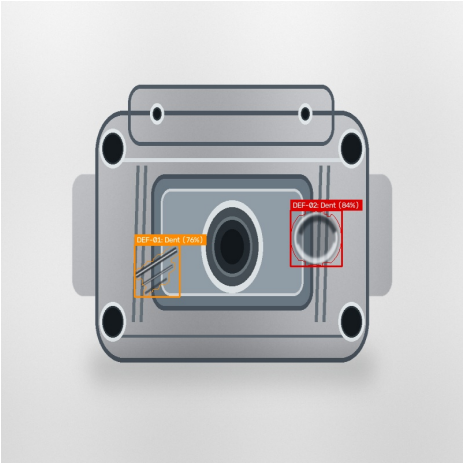


Figure D: Defect Highlighted & Annotated



Detected Defects & Anomaly Registry

Defect ID	Category	Severity	Score	Location (X, Y, W, H)	Area (% / px)	Description

DEF-01	**Dent**	**MEDIUM**	0.76	(231, 422) 78x90	0.51% (3271px)	Localized plastic deformation / depression (0.51% surface area).
DEF-02	**Dent**	**CRITICAL**	0.84	(501, 362) 89x97	1.05% (6733px)	Localized plastic deformation / depression (1.05% surface area).

AUTOMATED QUALITY ENGINEERING RECOMMENDATIONS:

- Verify whether surface dent impacts critical mating face sealing or bolt seat flatness.
- Check material handling conveyors and robotic gripper pad pressure to prevent impact indentation.

Generated by Smart Manufacturing AI – Computer Vision Quality Inspection Intelligence Suite. Certified for plant floor auditor sign-off.